Information Security (CS-3002)

Assignment # 3

Note: This is an individual assignment. Each student must submit their own original work. Collaborative or identical submissions will be treated as plagiarism.

Part 1: Email Security Implementation (10 Marks)

Objective:

To practically apply the security guidelines and protection mechanisms discussed in the provided Email Security document.

Instructions:

1. Carefully read the attached document: **"Email Security.pdf"**
2. Follow each instruction exactly as described in the document. step by step. use your real or dummy email account.
3. For each step performed, take clear *screenshots* and write a short caption explaining what you did (in your own words).
4. Compile all screenshots in order and save them in a single file named: Part1_EmailSecurity_[RollNo].pdf

Note: You must strictly follow the document as it is. No skipping or altering of steps. Marks will be awarded based on correctness, completeness, and clarity of screenshots.

Part 2: Research-Based Technical Investigation (10 Marks)

Paper: *"Ensemble Machine Learning Approach for Android Malware Classification Using Hybrid Features"* by Abdurrahman Pektaş and Tankut Acarman (2017)

Objective:

To analyze, critique, and apply understanding of hybrid malware classification through deeper reasoning beyond AI-generated text.

Important Note:

- Submissions will be checked through **AI-content detection tools**.
- Use **your own handwriting**; copy-pasted or LLM-generated answers will receive **zero marks**.
- Diagrams or handwritten analytical notes are mandatory for this part.

Q1. Conceptual Model Diagram (2 Marks)

Draw a detailed flowchart or conceptual diagram of the hybrid malware classification model discussed in the paper. Your diagram must show the entire flow from data collection to result evaluation and include both static and dynamic feature phases. Label each block using your own short explanations (not copied terms). You may draw by hand or digitally.

Q2. Analytical Explanation (2 Marks)

The authors compared **Random Forest**, **Logistic Regression**, and **Naive Bayes**. Explain using technical reasoning why **Random Forest** achieved the highest accuracy. Your answer must discuss the nature of hybrid features, imbalance of dataset, and adaptability of ensemble learners.

Q3. Cross-Domain Extension **(2 Marks)**

Imagine applying the same ensemble-based hybrid model to detect email phishing instead of Android malware.
Answer the following:

i. What would represent static and dynamic features in an email context?
ii. Identify two new security risks or challenges that might arise when transferring this ML approach from Android to email.

Q4. Experimental Design Critique **(2 Marks)**

The authors processed 3339 malware samples across 20 families using multiple virtual machines.

i. Identify one strength (e.g., scalability, realism, etc.) and one limitation (e.g., dataset imbalance, environment bias).
ii. Suggest one concrete improvement that could enhance the evaluation's fairness or robustness.

Q5. Innovative Enhancement **(2 Marks)**

If you were tasked to enhance the proposed hybrid malware detection model:

i. Propose one advanced AI or security method (e.g., federated learning, explainable ML, adversarial training, or blockchain logging).
ii. Explain how it could improve trust, privacy, or scalability in malware detection linking your explanation to CIA (Confidentiality, Integrity, Availability) principles.

Submission Format

- Combine Part 1 and Part 2 into a single PDF file:
  Assignment3_[Your RollNo].pdf
- Include your Name, Roll Number, and Section on the first page.
- Submit it on GCR.

Marking Rubric

Criterion	Description	Marks
Part 1 – Email Security	Screenshots correctly follow provided instructions, with proper sequence and captions	10
Part 2 – Technical Analysis	Original thought, depth of reasoning, correctness of diagram, conceptual understanding	10
Bonus (Optional)	Exceptional creativity in cross-domain analysis or security model redesign	+2